

3.2 Linear Momentum & Conservation

Question Paper

Course	CIEA Level Physics
Section	3. Dynamics
Topic	3.2 Linear Momentum & Conservation
Difficulty	Easy

Time allowed: 40

Score: /30

Percentage: /100

Question 1a

State the principle of conservation of linear momentum.

[2 marks]

Question 1b

A pellet of mass 20 g is fired from an air rifle with a momentum of 1.2 kg m s^{-1} .

Calculate the velocity of the bullet just after the rifle is fired.

[2 marks]

Question 1c

Use the principle of conservation of momentum to state:

(i)
the total momentum of the air rifle and the pellet,

[1]

(ii)
the recoil momentum of the air rifle.

[1]

[2 marks]

Question 1d

The pellet hits a target. It becomes embedded and takes 0.0025 s to come to rest.

(i)

State the relation between force and momentum.

[1]

(ii)

Calculate the average force needed to stop the pellet.

[3]

[4 marks]

Question 2a

A collision can be described as being elastic or inelastic when there are no external forces acting on the collision.

In Table 1.1, place a tick (✓) next to the quantities that are conserved in each type of collision

Table 1.1

Quantity	Elastic Collision	Inelastic Collision
Momentum		
Total Energy		
Kinetic Energy		

[3 marks]

Question 2b

A trolley **X** with mass of 400 g is moving on a track with a velocity of 1.2 m s^{-1} towards a stationary trolley **Y** with mass of 800 g as shown in Fig. 1.1.

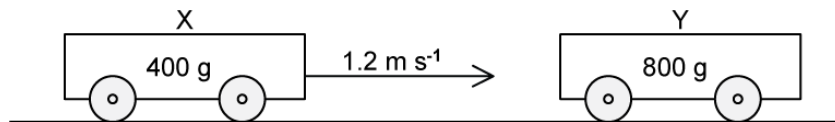


Fig. 1.1

(i)

Calculate the initial momentum of trolley **X**.

[2]

(ii)

Trolley **X** collides with trolley **Y** and subsequently, they stick together.

State the combined momentum of the trolleys after the collision.

[1]

[3 marks]

Question 2c

(i)

Calculate the velocity of the two trucks after the collision.

[3]

(ii)

State the direction of the velocity of the trolleys after the collision.

[1]

[4 marks]**Question 2d**

Explain why the collision cannot be described as elastic.

[1 mark]**Question 3a**

State the difference between an elastic and an inelastic collision.

[2 marks]

Question 3b

A ball collides with a wall at a velocity of 35 m s^{-1} and rebounds with velocity v . The ball has a mass of 0.025 kg .

The motion of the ball is shown in Fig. 1.1.

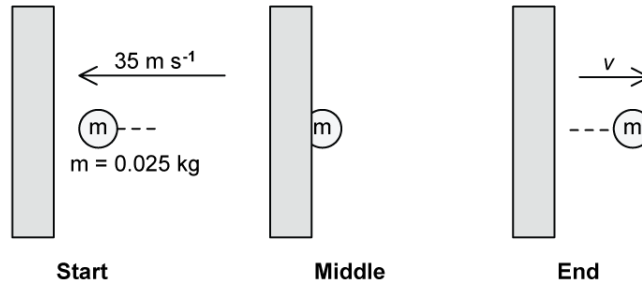


Fig. 1.1

Calculate the momentum of the ball at the start, before it hits the wall.

[2 marks]

Question 3c

The collision is perfectly elastic.

Determine

(i)
the rebound velocity of the ball

[1]

(ii)
the magnitude of the change in momentum of the ball.

[2]

[3 marks]

Question 3d

The ball is in contact with the wall for 0.1 s.

Calculate the force exerted on the ball by the wall.

[2 marks]

Model Answers: Easy

1a

(a) The principle of conservation of linear momentum states:

- (In a closed system) the total momentum / sum of momenta (before and after a collision) is constant; [1 mark]
- Provided no resultant / external forces act; [1 mark]

[Total: 2 marks]

1b

(b) Calculate the velocity of the pellet just after the air rifle is fired:

List the known quantities:

- Mass of pellet, $m = 20 \text{ g} = 0.020 \text{ kg}$
- Momentum of the pellet, $p = 1.2 \text{ kg m s}^{-1}$

Use the equation for momentum to calculate the final velocity:

- Momentum: $p = mv \Rightarrow v = \frac{p}{m}$ [1 mark]
- $v = \frac{1.2}{0.020}$
- $v = 60 \text{ m s}^{-1}$ [1 mark]

[Total: 2 marks]

1c

(c)

(i) State the total momentum of the air rifle and pellet:

- Conservation of momentum: Total momentum before = total momentum after
- Total momentum of air rifle and pellet = 0; [1 mark]

(ii) State the recoil momentum of the air rifle:

- Momentum of the pellet after firing, $p = 1.2 \text{ kg m s}^{-1}$
- (Hence) the recoil momentum of the air rifle = -1.2 kg m s^{-1} [1 mark]

[Total: 2 marks]

Remember that momentum is a **vector** quantity. This means it is important to consider the **direction** as well as the magnitude.

When the air rifle is fired it will recoil in the **opposite direction** to the bullet. To show that the rifle and the bullet are acting in opposite directions we use a **minus sign** when calculating the total momentum.

The total momentum **before** is when the bullet still hasn't been shot out or the rifle recoil. Therefore, it is 0.

1d

(d)

(i) The relation between force and momentum is:

- Force = rate of change of momentum **OR** $F = \frac{\Delta p}{\Delta t}$; [1 mark]

(ii) Calculate the average force needed to stop the pellet:

List the known quantities:

- Momentum of pellet before = 1.2 kg m s^{-1}
- Velocity of pellet after embedded = 0 m s^{-1}
- Time taken, $\Delta t = 0.0025 \text{ s}$

Determine the change in momentum:

- Change in momentum: $\Delta p = \text{momentum before} - \text{momentum after}$
- $\Delta p = 1.2 - 0 = 1.2 \text{ kg m s}^{-1}$ [1 mark]

Determine the average force on the pellet:

- $F = \frac{\Delta p}{\Delta t} = \frac{1.2}{0.0025}$ [1 mark]
- $F = 480 \text{ N}$ [1 mark]

[Total: 4 marks]

2a

(a)

A table that would score 3 marks looks like this:

Quantity	Elastic Collision	Inelastic Collision
Momentum	✓	✓
Total Energy	✓	✓
Kinetic Energy	✓	

- Momentum conserved in **both** elastic and inelastic collisions; [1 mark]
- Total energy conserved in **both** elastic and inelastic collisions; [1 mark]
- Kinetic energy conserved in elastic collisions **only**; [1 mark]

[Total: 3 marks]

In a closed system (i.e., no matter or mass moves in or out), momentum is **always** conserved during a collision, provided **no external force acts** on the system. The law of conservation of energy can never be violated, therefore the **total** energy before and after the collision must be conserved. Kinetic energy is only conserved in elastic collisions. During inelastic collisions some of the kinetic energy gets converted into other forms such as thermal energy.

2b

(a)

List the known quantities:

- Mass of **X** = 400 g = 0.4 kg
- Velocity of **X** = 1.2 m s^{-1}

(i) Calculate the initial momentum of trolley **X**:

- Momentum: $p = mv$ [1 mark]
- Momentum before: $p = 0.4 \times 1.2$
- Momentum before: $p = 0.48 \text{ kg m s}^{-1}$ [1 mark]

(ii) The combined momentum after the collision is:

- Momentum before the collision = momentum after the collision
- Momentum after: $p = 0.48 \text{ kg m s}^{-1}$ [1 mark]

[Total: 3 marks]

2c

(c)

(i) Calculate the velocity of the two trucks after the collision:

List the known quantities:

- Mass of **X** = 0.4 kg
- Mass of **Y** = 0.8 kg
- Momentum of the combined trucks, $p = 0.48 \text{ kg m s}^{-1}$

Determine the new mass when the trucks are stuck together:

- Total mass after collision = $(0.4 + 0.8) = 1.2 \text{ kg}$ [1 mark]

Rearrange the momentum equation and substitute in the values:

- $p = mv \Rightarrow v = \frac{p}{m}$
- Velocity after: $v = \frac{0.48}{1.2}$ [1 mark]
- Velocity after: $v = 0.4 \text{ m s}^{-1}$ [1 mark]

(ii) The direction of the final velocity will be:

- The same direction / to the right; [1 mark]

[Total: 4 marks]

Since velocity is a vector, and so is momentum, the positive or negative sign tells you its direction. In this scenario, the initial direction of trolley X has been taken as positive. Since the final momentum is positive, this implies the trolleys are moving in the same direction as trolley X at the beginning.

2d

(d) The collision is not elastic because:

- The final velocity / kinetic energy is less after the collision;

OR

The total kinetic energy before the collision is greater than the total kinetic energy after the collision.

- Mass of X = 0.4 kg
- Mass of Y = 0.8 kg
- Momentum of the combined trucks, $p = 0.48 \text{ kg m s}^{-1}$

Determine the new mass when the trucks are stuck together:

- Total mass after collision = $(0.4 + 0.8) = 1.2 \text{ kg}$ [1 mark]

Rearrange the momentum equation and substitute in the values:

- $p = mv \Rightarrow v = \frac{p}{m}$
- Velocity after: $v = \frac{0.48}{1.2}$ [1 mark]
- Velocity after: $v = 0.4 \text{ m s}^{-1}$ [1 mark]

(ii) The direction of the final velocity will be:

- The same direction / to the right; [1 mark]

[Total: 4 marks]

Since velocity is a vector, and so is momentum, the positive or negative sign tells you its direction. In this scenario, the initial direction of trolley X has been taken as positive. Since the final momentum is positive, this implies the trolleys are moving in the same direction as trolley X at the beginning.

2d

(d) The collision is not elastic because:

- The final velocity / kinetic energy is less after the collision;
OR
- The relative speed of separation is less than the relative speed of approach; [1 mark]

[Total: 1 mark]

If you're not sure how to calculate the relative speeds, it is simply:

- The relative speed of approach: $(u_X - u_Y) = (1.2 - 0) = 1.2 \text{ m s}^{-1}$

- The relative speed of separation: $v_{XY} = 0.4 \text{ m s}^{-1}$

It's useful to note that when two objects stick together after a collision, they can **never** be elastic

In fact when two objects stick together, it is said to be “perfectly” inelastic, because, in this scenario, the maximum amount of kinetic energy is dissipated.

You can think of this as the remaining energy wasted as heat or sound. If there was a collision of two cars where one drives into a stationary one, you would hear the sound of the collision.

3a

(a) The difference between an elastic and an inelastic collision is:

- In an elastic collision, the total kinetic energy is conserved **OR** relative speed of approach = relative speed of separation; [1 mark]
- In an inelastic collision, kinetic energy is not conserved / is lost **OR** relative speed of approach > relative speed of separation; [1 mark]

[Total: 2 marks]

3b

(b) Calculate the momentum of the ball at the start, before it hits the wall:

List the known quantities:

- Mass, $m = 0.025 \text{ kg}$
- Velocity, $v = 35 \text{ m s}^{-1}$

State the equation for momentum and calculate:

- Momentum: $p = mv$ [1 mark]
- Momentum: $p = 0.025 \times 35$
- Momentum: $p = 0.875 = 0.88 \text{ kg m s}^{-1}$ [1 mark]

[Total: 2 marks]

3c

(c)

List the known quantities:

- Mass, $m = 0.025 \text{ kg}$
- Initial momentum = $0.875 \text{ kg m s}^{-1}$
- Initial velocity = 35 m s^{-1}

(i) Determine the rebound velocity of the ball:

- For a perfectly elastic collision: relative speed of approach = relative speed of separation
- Hence, rebound (final) velocity: $v = -35 \text{ m s}^{-1}$ [1 mark]

(ii) Determine the change in momentum of the ball:

- Change in momentum: Momentum after – momentum before
- Change in momentum: $\Delta p = (0.025 \times -35) - (0.025 \times 35)$ **OR** 2×0.875 [1 mark]
- The magnitude of the change in momentum: $\Delta p = 1.75 = 1.8 \text{ kg m s}^{-1}$ [1 mark]

[Total: 3 marks]

The question demonstrates the importance of direction. Since momentum is a vector quantity, the direction influences the sign on the value.

The **magnitude** of the change in momentum means don't worry about the signs, the final answer should just be the **size**. However, the negative or positive signs are important to get to this stage!

If momentum was a scalar, you would say the 'change in momentum' was zero, as the magnitude of the momentum before and after didn't change. However, the fact that the momentum acts in the opposite direction afterwards means the magnitude of the total momentum is **twice** that of the momenta before and after the collision.

This very fact has crucial implications in the real world, such as in gas laws, where it can

be used to explain why gas pressure exists

3d

(d) Calculate the force exerted on the ball by the wall

List the known quantities:

- Change in momentum, $\Delta p = 1.75 \text{ kg m s}^{-1}$
- Change in time, $\Delta t = 0.1 \text{ s}$

Use the equation for force and momentum:

- $F = \frac{\Delta p}{\Delta t}$ [1 mark]

- $F = \frac{1.75}{0.1}$

- Force: $F = 17.5 = 18 \text{ N}$ [1 mark]

[Total: 2 marks]

